<u>***Electricity & Magnetism***</u>

We are surrounded by devices that depend on the physics of *electromagnetism*, which is the combination of *electric* and *magnetic* phenomena. When an ebonite rod is rubbed with fur (cat's hair) or cat skin, it gains a property due to which it attracts some other bodies like piece of paper. We know that the attraction between ebonite and paper is due to an *electric force*. If a certain type of stone (*a naturally occurring magnet*) is brought near bits of iron, the iron will jump to the stone. We know that the attraction between *magnet* and *iron* is due to a *magnetic force*.

<u>***Electrostatics***</u>

Electrostatics is the physics of stationary electric charge (*i.e. charge at rest*).

<u>***Concept of electric charge***</u>

Electric charge is a property that comes with particles wherever they exist. The vast amount of charge in an everyday object is usually hidden because the object contains equal amounts of the two kinds of charge: *positive* charge and *negative* charge. With an *equality* or *balance* of charge, the object is said to be *electrically neutral*; that is, it contains *no net charge*.

We say that an object is charged to indicate that it has a *charge imbalance*, or net *charge*. The imbalance is always much smaller than the total amounts of positive charge and negative charge contained in the object. Charges with the *same electrical sign repel each other*, and charges with *opposite electrical signs attract each other*.

<u>***Quantization of electric charge***</u>

Molecules, in general, are larger in size than atoms. A molecule usually contains a few atoms of either the same element or of different elements. The fundamental charge on the electron (or proton) is 1.6×10^{-19} *coulomb* denoted by *e*. Thus the charge on the proton is +*e* and on the electron is –*e*.

Since all matter in nature consists of an *integral* number of fundamental particles (atoms or molecules), the total charge q in any matter is integral multiple of e, $i.e.$,$q = ne$. In other words, we can only have charges $\pm e, \pm 2e, \pm 3e, \cdots\cdots, \pm ne$. This is known as *principle of quantization* of charge.

<u>***Coulomb's law***</u>

<u>Statement:</u> *The magnitude of the force between two point charges is directly proportional to the product of the magnitude of the charges and is inversely proportional to the square of the distance between them and acts on the line joining between the charges.*

<u>Explanation:</u> If q_1 and q_2 are the magnitude of two point charges and r is the distance between them, then the force

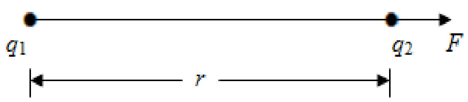

Fig. 1.

$$F \propto \frac{q_1 q_2}{r^2}$$

$$\therefore \quad F = k\frac{q_1 q_2}{r^2} \qquad (1)$$

The value of constant k depends on the choice of units of F, r and q and the nature of the medium in which the charges are situated.

When expressed in M.K.S. system, the Coulomb's law takes the form

$$F = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2} \quad (2)$$

Where, ε_0 is the permittivity of the free space.

In vector forms

$$\vec{F} = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^3}\vec{r} \quad (3)$$

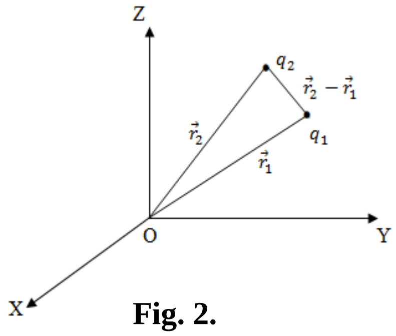

Fig. 2.

For radius vectors of charges q_1 and q_2 with respect to a fixed origin O

$$\vec{F} = \frac{q_1 q_2}{4\pi\varepsilon_0}\frac{\vec{r_2} - \vec{r_1}}{|\vec{r_2} - \vec{r_1}|^3} \quad (4)$$

Limitations of Coulomb's law: Coulomb's law is strictly applicable to charges at rest. When a charge is in motion forces other than electrostatic forces come into play and Coulomb's law cannot be used to measure the charges in motion.

Unit of charge: The S.I. unit of charge is *Coulomb*. It is defined from the basic unit of current in S.I. *i.e.*, an *ampere*.

A Coulomb is defined as the amount of charge that flows through a given cross-section of a wire in one second if there is a steady current of one ampere in the wire, i.e.,

$$q = It$$

If $I = 1\ amp\ and\ t = 1\ sec$ then $q = 1\ Coul$

Relative permittivity: Relative permittivity of a medium is denoted by ε_r and is defined as, *'the ratio of the permittivity of the medium to the permittivity of the free space'*. Mathematically

$$\varepsilon_r = \frac{\varepsilon}{\varepsilon_0}$$

Where, ε is the permittivity of the medium

ε_0 is the permittivity of the free space.

Mathematical problems

Problem-1: *The distance r between the electron and the proton in the hydrogen atom is about* 5.3×10^{-11} *meter? What are the magnitudes of*

(i) The electric force

(ii) The gravitational force between these two particles.

Solution: From Coulomb's law,

$$F_e = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2}$$

$$Or \qquad F_e = \frac{9 \times 10^9 \times (1.6 \times 10^{-19})^2}{(5.3 \times 10^{-11})^2}$$

$$\therefore \qquad F_e = 8.1 \times 10^{-8}\ N\ (\boldsymbol{Ans})$$

The gravitational force is given by

$$F_g = G\frac{m_1 m_2}{r^2}$$

$$Or \quad F_g = \frac{6.67 \times 10^{-11} \times 9.1 \times 10^{-31} \times 1.7 \times 10^{-27}}{(5.3 \times 10^{-11})^2}$$

$$\therefore \quad F_g = 3.68 \times 10^{-47}\ N\ (\boldsymbol{Ans})$$

Here,

$q_1 = q_2 = 1.6 \times 10^{-19}\ C$

$r = 5.3 \times 10^{-11}\ m$

$\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9\ N - m^2/C^2$

$G = 6.67 \times 10^{-11} Nm^2kg^{-2}$

$m_1 = 9.1 \times 10^{-31} kg$

$m_2 = 1.7 \times 10^{-27} kg$

$F_e = ?$

$F_g = ?$

Problem-2: *Calculate the repulsive Coulomb force exist between two protons in a nucleus of iron. Assume a separation of* $4 \times 10^{-15}\ m$.

Solution: From Coulomb's law,

$$F = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2}$$

$$Or \qquad F = \frac{9 \times 10^9 \times (1.6 \times 10^{-19})^2}{(4.0 \times 10^{-15})^2}$$

$$\therefore \qquad F = 14\ N\ (\boldsymbol{Ans})$$

Here,

$q_1 = q_2 = 1.6 \times 10^{-19}\ C$

$r = 4.0 \times 10^{-15}\ m$

$\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9\ N - m^2/C^2$

$F = ?$

Problem-3: *A negative charge of* 10^{-6} *is situated in air at the origin of a rectangular co-ordinate system. A second negative point charge of* 10^{-4} *is situated on the positive X-axis at a distance of 50 cm from the origin. What is the force on the second charge?*

Solution: From Coulomb's law,

$$F = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2}$$

$$Or \qquad F = \frac{9 \times 10^9 \times (-1 \times 10^{-6}) \times (-1 \times 10^{-4})}{(0.5\ )^2}$$

$$\therefore \qquad F = 3.6\ N\ (\boldsymbol{Ans})$$

Thus there is a force of 3.6 N in the positive X-direction on the second charge. ***(Ans)***

Here,

$q_1 = -1 \times 10^{-6}\ C$

$q_2 = -1 \times 10^{-4}\ C$

$r = 50\ cm = 0.5\ m$

$\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9\ N - m^2/C^2$

$F = ?$